Agence spatiale canadienne | Canadian Space Agency

Stratospheric Balloons Program

Science Timmins Experiment Description

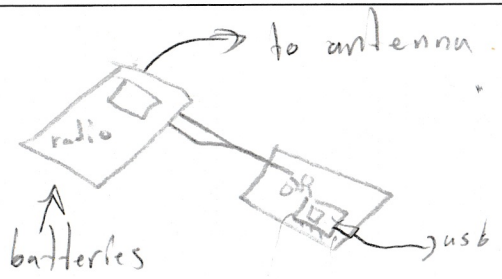
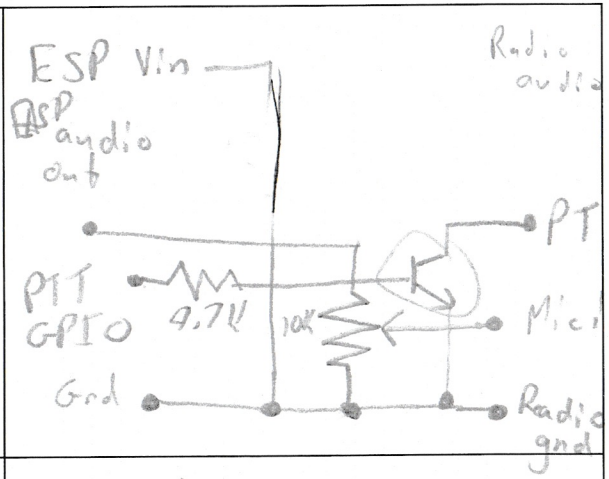
2025 Campaign – Phase I

Please fill out this table to the best of your ability. You don't need to know all the answers at the start, it can be updated later (there will be three phases).

Student Name(s)		Gabriel Jacques
Experiment Name *Come up with a name for your experiment, be creative! Think about what the experiment will do, what it wants to measure, what it will look like, etc. This can be an abbreviation, an acronym, or a word.* *For example, an experiment that wants to **ME**asure **O**xygen and **W**ater vapour could be named "MEOW".*		
Experiment Objective(s) *What do you want to achieve with the experiment? What specific things will it measure or demonstrate to achieve its objective(s)?* *For example, the "MEOW" experiment's goal is to measure Oxygen, but there are multiple ways to do that, such as using an Oxygen sensor to measure the Parts Per Million (ppm) of Oxygen in the local atmosphere around the balloon.*		Offer a means to transmit live information using an amateur radio. +Possibly recieve commands.
Experiment Description *Describe the experiment in each of these areas. If there is no electrical or software, you can write "Not Applicable".*	**Electrical** *If experiment uses electricity, which parts require power, and at what Voltage (Volts)? How much power will they consume (Watts)?*	ESP-32 (powered with USB)
	Mechanical *What does the experiment physically look like? Where will the components go, how will they be attached? Will the experiment fit in half of a payload pod, or require two halves? What is the expected mass of the experiment?*	ESP-32 connected to voltage regulating circuit connected to Baofeng Half-payload
	Software	

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	Does the experiment involve software? What On-Board Computer (OBC) will be used, and what will the code do?	ESP32 → Circuit → Baofeng → Receiver dongle → Direwolf → python → SQL → Php website data
	Components *What components will be used in the payload? Have you selected a specific kind, or model?*	Baofeng-UV-5R ESP-32 APRS cable 2x 10kΩ pots 0.1μF capacitor Breadboard? 4.7kΩ resistor NPN transistor ? or 1 ? or 0

Experiment Photos *"A picture is worth a thousand words," please add pictures of the experiment when you can. These can be sketches, 3D models, flow charts, block diagrams, or photos of the experiment. They can show just a component/portion of the experiment, or the whole thing. Feel free to add more pages if needed!*		

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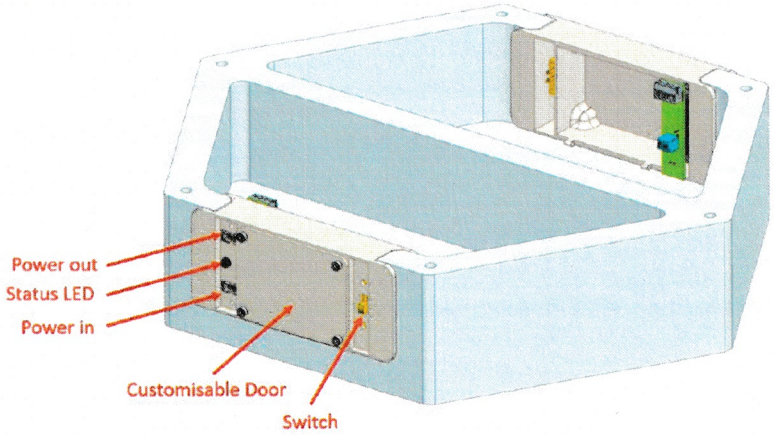
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Background Information

Payload Pod

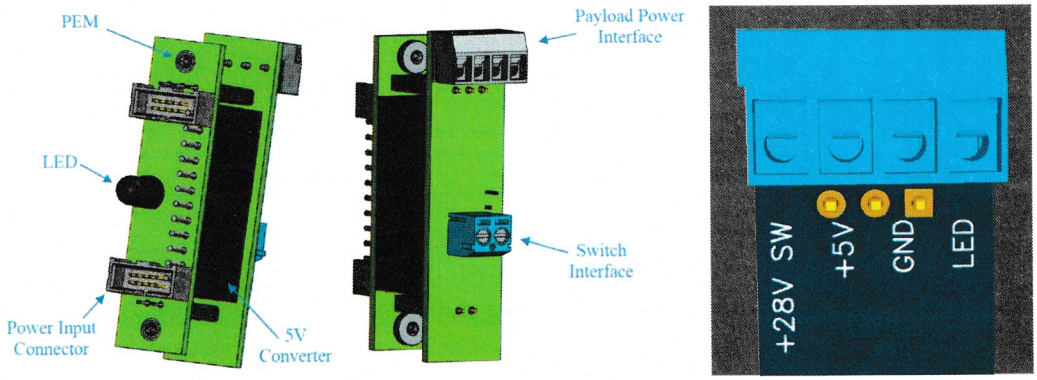
The payload pod is a generic foam enclosure designed to contain experiments and provide thermal insulation against the cold stratospheric environment and intense solar radiation. Each pod has two compartments, and an experiment can use one or both compartments. The central and outer walls are needed to ensure the pods are not crushed when they are vertically stacked, and the foam enclosure cannot be modified, except to add a few small holes for wire pass-through between the two chambers. The internal volume of the pod is a trapezoidal shape that is 120mm wide, has side lengths of 150mm and 290mm, and a height of 57mm.

Each compartment has one payload pod interface. This interface connects the experiment inside the pod to the CSA power subsystem underneath the pods. It also has a switch that can be used to turn power on or off to the experiment and a status LED that can be programmed to indicate the status of the experiment. The interface also has a customisable door to the payload that is attached with four bolts. As needed, this door can be modified to incorporate features, such as a view port for a camera, needed by the experiment.



Power System

The CSA power system has a capacity of 60 Whr, shared among all the payloads operating on it. This power system connects to each payload pod via the power interface board (shown below), which is part of the pod interface (shown above). On the inside of this interface (the experiment side), there is a Payload Power Interface and Switch Interface, as shown below. Both these interfaces are screw terminals.



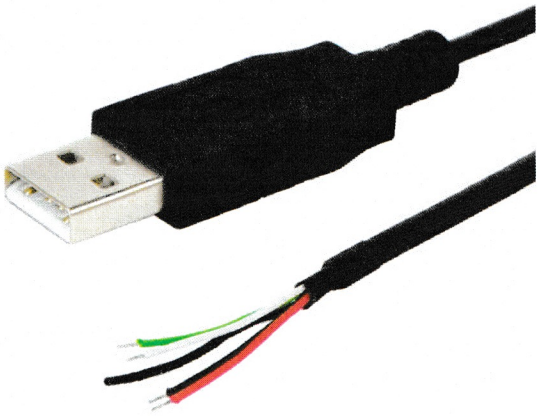
The Payload Power Interface, shown above, provides four channels:

- *"+28V SW", a 28 V nominal power channel that ranges from 33.6 V to 24 V and is limited to 2.7 A,*
- *"+5V", a +5 V power channel that is limited to 3 A,*
- *"GND", an electrical ground for the experiment, and*

- *"LED", the positive voltage input to the LED*

If a payload requires a voltage other than +5V, it will require voltage conversion using either a switched DC-DC converter or a regulator. A DC-DC converter can generate higher voltages, lower voltages, and even negative voltages. A regulator can only generate lower voltages and negative voltages, but will dissipate much more heat, which might be problematic in the stratosphere.

If the payload is using a controller such as an Arduino or a Raspberry Pi that can be supplied +5V through a USB port, then be prepared to modify a USB cable by cutting it and stripping the jacket to expose the red and black supply cables in it. Red is positive (connects to +5V), black is negative (connects to GND).



Make sure to select the proper USB connector for the controller (micro-usb, usb-c, etc.)



Micro-USB



USB-C

Cables with exposed wires can also be purchased from distributors such as Digikey, Adafruit or Sparkfun.

The LED connection is there so that payloads can have a visual indicator that the payload is powered. At the very least, it can be connected directly to the +5V screw terminal contact. By doing this, the LED will light up once the power switch is turned on. A better idea is to connect it to a GPIO (if available) on the controller board. By toggling that GPIO on and off, the LED can be made to blink, and this is a very useful indicator that the controller software is running.